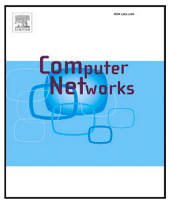
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Joint service caching, computation offloading and resource allocation for dual-layer aerial Internet of Things

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ABSTRACT

The exponential growth of Internet of Things devices has triggered an unprecedented surge in mobile data traffic, posing significant challenges for latency-sensitive services. Mobile Edge Computing (MEC) has emerged as a promising solution by decentralizing computation and caching resources to the network edge. However, traditional terrestrial MEC systems struggle with limited coverage and flexibility. To overcome these issues, this paper proposes a novel dual-layer aerial MEC architecture, where multiple Unmanned Aerial Vehicles (UAVs) provide computation and caching support for resource-constrained terminal devices, and a high-altitude platform serves as a central hub for long-term service storage and retrieval. The system aims to minimize total latency by jointly optimizing service caching, task offloading, resource allocation, and 3D UAV deployment, formulated as a mixed-integer nonlinear programming problem and efficiently solved using an iterative algorithm based on linear relaxation and successive convex approximation. Simulation results demonstrate that the proposed scheme converges quickly across different scales and outperforms all baselines with minimal runtime increase, reducing total latency by 42.86% compared to the random UAV deployment.

1. Introduction

The rapid development of the Internet of Things (IoT) and mobile communication technologies has triggered a dramatic surge in mobile data traffic [1,2]. As IoT devices proliferate across diverse sectors such as industrial automation, smart cities, telemedicine, and autonomous vehicles, the demand for low-latency and high-throughput network services becomes increasingly stringent [3]. Despite cloud computing architectures providing extensive computing and storage resources, their inherent centralization results in substantial transmission latency, which poses challenges in meeting these stringent demands [4,5]. To overcome these hurdles, Mobile Edge Computing (MEC) has emerged as a transformative paradigm by pushing multi-dimensional resources to the network edge, thereby reducing core network congestion and improving response times [6,7].

However, traditional terrestrial MEC systems face geographic coverage and spatial flexibility limitations, as their fixed infrastructure struggles to efficiently support emergency or remote environments. In contrast, aerial platforms such as Unmanned Aerial Vehicles (UAVs) and High-Altitude Platforms (HAPs) provide a more flexible and resilient alternative for extending MEC services [8,9]. With their rapid deployment and high mobility in three-dimensional (3D) space, UAVs are ideal for delivering immediate connectivity to localized regions [10]. HAPs, on the other hand, offer stable, wide coverage from the stratosphere, supporting persistent communication over broad regions [11]. Integrating UAVs and HAPs in a dual-layer MEC architecture balances short-term responsiveness with long-term stability, providing an adaptive and scalable solution for areas with limited or unreliable terrestrial infrastructure.

Service caching is an advanced technique for improving the efficiency of distributed systems by storing frequently used computation services, such as machine learning models, data processing pipelines, or application logic, closer to the Terminal Devices (TDs) [12,13]. Unlike traditional content caching, which focuses on static data, service caching ensures that essential services are readily available at edge nodes. This allows TDs to offload computation tasks and receive faster processing without frequent reliance on distant cloud servers [14,15]. As a result, service caching is increasingly recognized as a key enabler for applications requiring real-time responses, such as IoT, augmented reality, and autonomous vehicles [16]. However, service caching introduces new challenges, such as determining which services to cache, when to update or evict, and how to handle service requests' dynamic and heterogeneous nature.

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1.1. Related works

UAV-assisted MEC: UAV-assisted MEC systems have gained significant attention in recent years, with extensive research focusing on optimizing task offloading, resource allocation, and UAV deployment [17–21]. For instance, [17] proposed a multi-UAV task offloading and resource allocation framework that minimizes latency and energy consumption. Similarly, [18] developed an optimization scheme for joint task offloading and UAV trajectory design, aiming to maximize computation offloading energy efficiency. [19] focused on minimizing energy consumption in a multi-UAV edge computing system by optimizing UAV trajectories, offloading decisions, and resource allocation. [20] explored cooperative multi-UAV systems to enhance resource utilization, while [21] optimized dynamic scheduling and task allocation of UAVs to balance energy consumption and task completion time. However, these works focus solely on single-layer UAV networks and overlook the potential synergies between UAVs and HAPs, which could enable a more scalable multi-layer network infrastructure. Additionally, many studies assume fixed UAV altitudes, which limits the benefit of leveraging the 3D spatial flexibility of UAVs.

HAP integration in multi-layer MEC: Recent research has explored the integration of UAVs and HAPs in dual-layer systems, highlighting their benefits to enhance MEC performance [22–26]. [22] proposed a hierarchical aerial computing system that integrates UAVs and an HAP to maximize the number of computed tasks through optimized resource allocation and task offloading. [23] explored a dual-layer system with UAVs and HAPs, focusing on minimizing processing costs by improving task offloading efficiency. [24] emphasized energy-efficient task offloading and resource allocation in an air-ground integrated system, aiming to reduce energy consumption. Moreover, [25] investigated an HAP-supported MEC architecture designed for remote areas, focusing on minimizing latency and energy costs. Although these studies demonstrate the advantages of integrating HAPs over single UAV networks, they often assume that all services are readily available from remote servers, overlooking the additional latency caused when services are not cached locally at the edge. Addressing this issue is crucial for unlocking the full potential of dual-layer aerial MEC networks, especially in latency-sensitive applications.

Service caching strategies in MEC: Service caching is vital for reducing latency in MEC systems by storing frequently accessed services at the edge. Recent studies have focused on optimizing caching strategies for edge computing environments [26–29]. [26] proposed a joint service placement and task offloading strategy in an air-ground integrated network to minimize the long-term average weighted sum of latency and economic expenditure. [27] explored a cooperative MEC system, where service caching, resource allocation, and computation offloading were jointly optimized to minimize system cost. [28] examined task offloading and caching strategies in an MEC-enabled Internet of Vehicles network, aiming to minimize average execution latency. Additionally, [29] investigated cooperative computation and cache scheduling in UAV-enabled MEC networks, focusing on minimizing UAV endurance time while ensuring efficient resource use. Despite these advances, many works are limited to terrestrial or single-layer systems and overlook the potential of integrating service caching across both UAVs and HAPs in dual-layer architectures. To maximize the benefits of caching in such systems, it is essential to jointly optimize caching decisions with task offloading, resource allocation, and 3D UAV deployment.

1.2. Motivations and contributions

Based on the analysis above, several key challenges must be addressed in dual-layer aerial MEC systems.

- **Limited cache capacity of UAVs.** Each UAV has constrained storage capability, which limits the number of services it can cache. As a result, service caching strategies must be carefully designed to maximize the utility of the available cache while minimizing service retrieval latency from the HAP.
- **Task offloading decision.** TDs need to offload tasks to UAVs to conserve their own energy and processing power. However, determining when and where to offload tasks, based on UAV workload, service availability, and communication conditions, is a complex decision that directly affects system performance.
- **Bandwidth allocation.** UAVs must serve multiple TDs and communicate with the HAP using limited bandwidth. Allocating bandwidth efficiently to avoid communication bottlenecks is a critical challenge, especially when multiple TDs attempt to offload tasks to the same UAV simultaneously.
- **Coupling of multiple variables.** The interaction between service caching, task offloading, bandwidth allocation, and UAV deployment creates a tightly coupled system. Optimizing any one of these factors in isolation can lead to suboptimal performance in others. This tight coupling makes the optimization problem complex and computationally challenging.

In this paper, we propose a dual-layer aerial MEC architecture where multiple UAVs and an HAP collaborate to provide computation and caching support for resource-constrained terrestrial TDs. The HAP is a central data hub for all services, while UAVs selectively cache certain services to minimize backhaul transmission latency. TDs are fully covered by UAVs, while UAVs operate within the coverage of the HAP. The objective is to jointly optimize service caching, computation offloading, and resource allocation to minimize total latency, subject to caching capacity, maximum workload, and communication resources.

The main contributions are summarized as follows.

- **Dual-layer aerial MEC architecture.** We propose a novel MEC system model integrating UAVs and an HAP. UAVs offer computation and caching services, while the HAP is a central data hub. TDs either process tasks locally or offload them to UAVs, which execute the tasks immediately if the required service is cached, or retrieve the service from the HAP before executing.
- **Unified optimization scheme.** We develop a unified scheme that jointly optimizes service caching, task offloading, bandwidth allocation, and 3D multi-UAV deployment. The latency minimization problem is formulated as a Mixed-Integer Nonlinear Programming (MINLP), and we propose an iterative algorithm based on linear relaxation and Successive Convex Approximation (SCA) techniques.
- **Low-complexity iterative algorithm.** We present a low-complexity algorithm that decomposes the MINLP problem into subproblems, solving them iteratively to approach the global optimum. Given the limited cache capacity of UAVs, we develop a priority-based caching strategy that selects services based on their contribution to reducing total latency.
- **Performance evaluation.** Simulation results demonstrate that the proposed scheme consistently outperforms other baselines, including nearest offloading, UAVs without caching, K-Means deployment, and equal bandwidth allocation, with only a minimal increase in runtime. Additionally, compared to the random UAV deployment, our scheme achieves a 42.86% reduction in total latency.

The remainder of this paper is organized as follows. Section 2 describes the system model and problem formulation. In Section 3, we propose an iterative algorithm for solving the optimization problem. Section 4 provides the complexity and convergence analysis. Section 5 presents the simulation results and discussions. Finally, Section 6 concludes the paper. The main notations used in this paper are summarized in Table 1.

Table 1
Notation list.

Notations	Descriptions
$K/\mathcal{K}$	Number/set of TDs
$M/\mathcal{M}$	Number/set of UAVs
$I/\mathcal{I}$	Number/set of services
h_m/h_{H}	Altitude of UAV m/HAP
D_{H}	Storage capacity of HAP
D_m	Cache capacity of UAV m
d_i	Data size of service i
$l_{k,i}$	Data size of task $L_{k,i}$
$c_{k,i}$	Required CPU cycles to complete task $L_{k,i}$
$\mathbf{w}_k/\mathbf{w}_{\mathrm{H}}$	Location of TD k/HAP
$\mathbf{q}_m$	Location of UAV m
$h_{k,m}$	Channel power gain between TD k and UAV m
$h_{\mathrm{H},m}$	Channel power gain between HAP and UAV m
$R_{k,m}$	Data rate from TD k to UAV m
$R_{\mathrm{H},m}$	Data rate from HAP to UAV m
$\alpha_{m,i}$	Caching decision for service i on UAV m
$\beta_{k,m}$	Task offloading decision for TD k
$\chi_{k,m}$	Fraction of bandwidth B_{U} allocated to TD k
β_0	Channel power gain at a distance of $d_0 = 1$ m
σ^2	Noise power at receivers
G_0	Total directional antenna gain
f_k^{loc}	Local computation capacity of TD k
$f_{k,m}$	Computation resources allocated by UAV m to TD k
p_k/p_{H}	Transmit power of TD k/HAP
$N_{\max}$	Maximum workload of each UAV
T_k^{loc}	Local computation latency of task $L_{k,i}$ at TD k
$T_{k,m}^{\mathrm{U}}$	Edge computation latency of task $L_{k,i}$ at UAV m
T_k	Total latency to complete task $L_{k,i}$

Notations: Vectors, matrices, and sets are denoted by bold lowercase letters **a**, bold uppercase letters **A**, and calligraphic uppercase letters $\mathcal{A}$, respectively. For a vector **a**, its transpose is denoted as $\mathbf{a}^{\mathrm{T}}$, and its Euclidean norm is represented by $|\mathbf{a}|$. The notation $|\cdot|$ refers to the absolute value.

2. System model and problem formulation

2.1. Network model

Consider a dual-layer aerial MEC architecture integrating communication, computation, and caching, as illustrated in Fig. 1. This architecture consists of one HAP, M rotary-wing UAVs equipped with edge servers, and K ground TDs. The sets of TDs and UAVs are denoted by $k \in \mathcal{K} = \{1, 2, \ldots, K\}$ and $m \in \mathcal{M} = \{1, 2, \ldots, M\}$, respectively. Each TD has an indivisible computation-intensive task requiring a specific service. The HAP serves as a central data hub storing all services, denoted by the set $i \in \mathcal{I} = \{1, 2, \ldots, I\}$, where the size of each service is d_i (in bits). The HAP has sufficient storage capacity, D_{H}, to store all services, i.e., $\sum_{i=1}^{I} d_i \le D_{\mathrm{H}}$, whereas each UAV has a limited cache capacity, D_m, that can only accommodate a subset of these services [30,31]. If the computation task of TD k requires service i, it is represented as $L_{k,i} \triangleq \{l_{k,i}, c_{k,i}\}$, where $l_{k,i}$ denotes the data size of the task, and $c_{k,i}$ represents the required CPU cycles to complete the task. The system workflow is depicted in Fig. 2 : (1) TD k offloads task $L_{k,i}$ to its associated UAV m. (2) UAV m checks whether the required service i is cached. If cached, UAV m processes the task directly; otherwise, it downloads service i from the HAP before processing. (3) Once task $L_{k,i}$ is completed, UAV m returns the computation result to TD k. Since the output data size is typically much smaller than the input data, phase (3) is often negligible.

2.2. Communication model

TD-to-UAV (T2U) and HAP-to-UAV (H2U) communications use different frequency bands to avoid cross-layer interference, with each UAV allocated a dedicated B_{H} for H2U communication. Each UAV employs

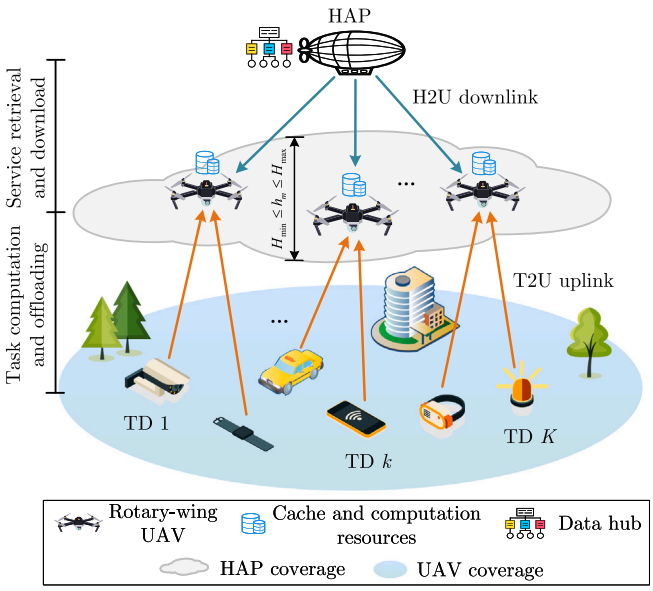


Fig. 1. Dual-layer aerial MEC architecture integrating communication, computation, and caching.

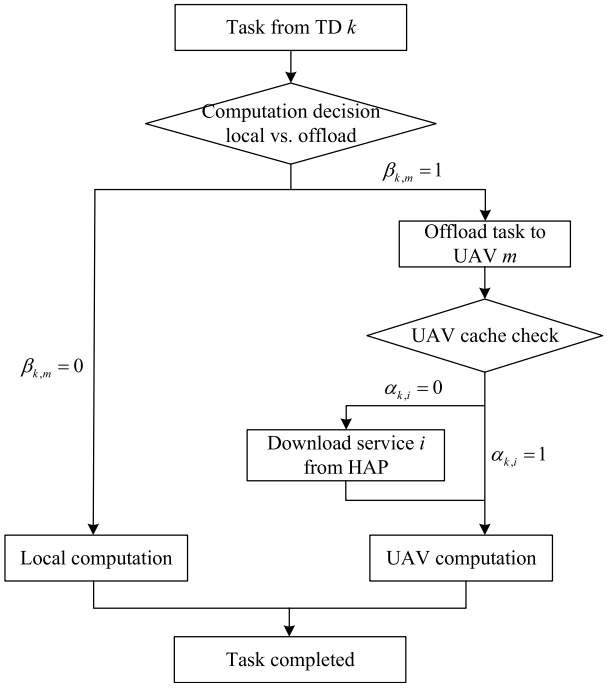


Fig. 2. Workflow of dual-layer aerial MEC system.

orthogonal frequency division multiple access to serve multiple TDs, and non-overlapping channels B_{U} are assigned to different UAVs to prevent co-channel interference [32,33].

Without loss of generality, a 3D Cartesian coordinate system is constructed. All TDs are fixed on the ground, with the location of TD k denoted as $\mathbf{w}_k = (x_k, y_k, 0)^{\mathrm{T}}$. Assume UAV m and the HAP hover at altitudes h_m and h_{H}, respectively, with their locations represented by $\mathbf{q}_m = (x_m, y_m, h_m)^{\mathrm{T}}$ and $\mathbf{w}_{\mathrm{H}} = (x_{\mathrm{H}}, y_{\mathrm{H}}, h_{\mathrm{H}})^{\mathrm{T}}$. Note that the locations of the TDs and the HAP are known, while the static 3D coordinates of each UAV are to be optimized. According to the Euclidean formula, the

distances between TD k and UAV m, and between the HAP and UAV m, are given by

$$d_{k,m} = \sqrt{\left(x_m - x_k\right)^2 + \left(y_m - y_k\right)^2 + h_m^2}, \forall k, m, \tag{1a}$$

$$d_{\mathrm{H},m} = \sqrt{\left(x_{\mathrm{H}} - x_m\right)^2 + \left(y_{\mathrm{H}} - y_m\right)^2 + \left(h_{\mathrm{H}} - h_m\right)^2}, \forall m. \tag{1b}$$

As UAVs hover at varying altitudes, signal propagation may be obstructed by buildings, trees, and other obstacles. Similar to [17–19], the channel power gain between TD k and UAV m is modeled by combining Rayleigh fading and Rician fading, expressed as

$$h_{k,m} = \frac{\beta_0}{d_{k,m}^{\alpha}} \left| \sqrt{\frac{K_{k,m}}{K_{k,m}+1}} g + \sqrt{\frac{1}{K_{k,m}+1}} \tilde{g} \right|^2, \forall k, m, \tag{2}$$

where $\alpha \geq 2$ is the path loss exponent, β_0 is the channel power gain at the reference distance $d_0 = 1$ m, $K_{k,m}$ is the Rician factor, g is the deterministic component of the Line-of-Sight (LoS) path with $|g| = 1$, and $\tilde{g}$ is the stochastic component of the non-LoS path, which follows a circularly symmetric complex Gaussian distribution with zero mean and unit variance.

Since the HAP and UAVs are located in the stratosphere and lower airspace, respectively, the H2U link is predominantly characterized by LoS propagation [24,25,34]. Therefore, considering free-space path loss and small-scale fading, the link between the HAP and UAV m is modeled as

$$h_{\mathrm{H},m} = G_0 \left(\frac{c}{4\pi d_{\mathrm{H},m} f_c} \right) \left| g_{\mathrm{H},m} \right|^2, \forall m, \tag{3}$$

where G_0 is the total directional antenna gain, c is the speed of light, f_c is the carrier frequency, and $g_{\mathrm{H},m}$ is the normalized small-scale fading coefficient.

Let B_{U} denote the bandwidth assigned to each UAV for communication with its associated TDs. Thus, the data rate (in bps/Hz) from TD k to UAV m is defined as

$$R_{k,m} = \chi_{k,m} B_{\mathrm{U}} \log_2 \left(1 + \frac{p_k h_{k,m}}{\sigma^2} \right), \forall k, m, \tag{4}$$

where p_k is the transmit power of TD k, σ^2 is the power of the additive white Gaussian noise at the receiver, and $\chi_{k,m} \in [0, 1]$ denotes the fraction of bandwidth B_{U} allocated by UAV m to TD k.

Let the transmit power of the HAP be p_{H}. The data rate from the HAP to UAV m is calculated as

$$R_{\mathrm{H},m} = B_{\mathrm{H}} \log_2 \left(1 + \frac{p_{\mathrm{H}} h_{\mathrm{H},m}}{\sigma^2} \right), \forall m, \tag{5}$$

where B_{H} represents the bandwidth assigned to each UAV for communication with the HAP.

2.3. Service caching model

Define a binary variable $\alpha_{m,i} \in \{0, 1\}$ to indicate the service caching decision. If service i is cached in UAV m, $\alpha_{m,i} = 1$; otherwise, $\alpha_{m,i} = 0$. The cache capacity constraint for each UAV is

$$\sum_{i=1}^{I} \alpha_{m,i} d_i \leq D_m, \forall m. \tag{6}$$

2.4. Task execution model

Each task can be computed locally at the TD or offloaded to one of the UAVs, forming $M + 1$ possible computation nodes represented by the set $\mathcal{M}' = \{0, 1, \ldots, M\}$. Define a binary variable $\beta_{k,m} \in \{0, 1\}$ for $m \in \mathcal{M}'$ to indicate the task offloading decision for TD k. If TD k offloads task $L_{k,i}$ to UAV m, then $\beta_{k,m} = 1$; otherwise, $\beta_{k,m} = 0$. Specifically, $\beta_{k,0} = 1$ indicates that the task is computed locally at TD k. Thus, all task offloading decisions must subject to the following constraint

$$\sum_{m=0}^{M} \beta_{k,m} = 1, \forall k. \tag{7}$$

Let the maximum workload of each UAV be $N_{\max}$. To prevent UAVs from being overloaded, the following constraint must be satisfied

$$\sum_{k=1}^{K} \beta_{k,m} \leq N_{\max}, \forall m. \tag{8}$$

2.4.1. Local computation

If TD k computes its task $L_{k,i}$ locally, the local computation latency is expressed as

$$T_k^{\mathrm{loc}} = \frac{c_{k,i}}{f_k^{\mathrm{loc}}}, \forall k, \tag{9}$$

where f_k^{loc} is the computation capacity of TD k. Services are typically pre-installed on the TDs, as their tasks are repetitive and predictable. Thus, we omit service backhauling from the HAP to the TDs. Future work could explore dynamic provisioning for less predictable tasks [15,26].

2.4.2. Edge computation at UAV

If TD k offloads task $L_{k,i}$ to UAV m for execution, the transmission latency from TD k to UAV m is given by

$$t_{k,m}^{\mathrm{off}} = \frac{l_{k,i}}{R_{k,m}}, \forall k, m. \tag{10}$$

Next, UAV m checks whether the service i is available in its cache. If the service is found, UAV m proceeds to compute the task. The computation latency at UAV m is expressed as

$$t_{k,m}^{\mathrm{comp}} = \frac{c_{k,i}}{f_{k,m}}, \forall k, m, \tag{11}$$

where $f_{k,m}$ represents the computation resources allocated by UAV m to TD k. If the service is not found in the cache, UAV m must download the required data from the HAP before computation. The transmission latency from the HAP to UAV m is expressed as

$$t_{\mathrm{H},m}^{\mathrm{down}} = \frac{d_i}{R_{\mathrm{H},m}}, \forall k, m. \tag{12}$$

Therefore, the edge computation latency of task $L_{k,i}$ at UAV m is calculated as

$$T_{k,m}^{\mathrm{edge}} = t_{k,m}^{\mathrm{off}} + t_{k,m}^{\mathrm{comp}} + \left(1 - \alpha_{m,i}\right) t_{\mathrm{H},m}^{\mathrm{down}}, \forall k, m. \tag{13}$$

In summary, the total latency to complete $L_{k,i}$ is the sum of the local and edge computation latencies, calculated as

$$T_k = \beta_{k,0} T_k^{\mathrm{loc}} + \sum_{m=1}^{M} \beta_{k,m} T_{k,m}^{\mathrm{edge}}, \forall k. \tag{14}$$

2.5. Problem formulation

Our objective is to minimize the total latency for all TDs by jointly optimizing the task offloading decisions $\mathbf{B} \triangleq \{\beta_{k,m}, \forall k, m\}$, bandwidth allocation $\mathbf{X} \triangleq \{\chi_{k,m}, \forall k, m\}$, 3D UAV deployment $\mathbf{Q} \triangleq \{\mathbf{q}_m, \forall m\}$, and service caching decisions $\mathbf{A} \triangleq \{\alpha_{m,i}, \forall m, i\}$, subject to UAV maximum workload, communication resources, and cache capacity constraints. The joint optimization problem is mathematically formulated as

$$(\mathrm{P1}): \quad \min T_{\mathrm{total}} \tag{15a}$$

$$\text{s.t.} \quad \sum_{k=1}^{K} \beta_{k,m} \chi_{k,m} \leq 1, \forall m, \tag{15b}$$

$$\sum_{i=1}^{I} \alpha_{m,i} d_i \leq D_m, \forall m, \tag{15c}$$

$$\sum_{m=0}^{M} \beta_{k,m} = 1, \forall k, \tag{15d}$$

$$\sum_{k=1}^{K} \beta_{k,m} \leq N_{\max}, \forall m, \tag{15e}$$

$$\chi_{k,m} \in [0,1], \forall k, m, \tag{15f}$$

$$\alpha_{m,i} \in \{0,1\}, \forall k, i, \tag{15g}$$

$$\beta_{k,m} \in \{0,1\}, \forall k, m, \tag{15h}$$

$$X_{\min} \le x_m \le X_{\max}, \forall m, \tag{15i}$$

$$Y_{\min} \le y_m \le Y_{\max}, \forall m, \tag{15j}$$

$$H_{\min} \le h_m \le H_{\max}, \forall m, \tag{15k}$$

where $T_{\text{total}} = \sum_{k=1}^{K} \left(\beta_{k,0} T_k^{\text{loc}} + \sum_{m=1}^{M} \beta_{k,m} T_{k,m}^{\text{edge}} \right)$, $X_{\min}$, $Y_{\min}$, and $H_{\min}$ represent the minimum allowed coordinates in the x, y, and z dimensions for the UAVs, while $X_{\max}$, $Y_{\max}$, and $H_{\max}$ correspond to the maximum allowed coordinates in these dimensions. (15b) ensures that the total bandwidth allocated by a UAV to TDs does not exceed its available communication bandwidth. (15c) ensures that the amount of data cached on each UAV does not exceed its cache capacity. (15d) enforces that each task must select exactly one computation node for execution. (15e) prevents UAVs from being overloaded with tasks. (15f) stipulates that the bandwidth allocation ratio is a real number between 0 and 1. (15g) and (15h) indicate binary decisions for service caching and task offloading, respectively. (15i) and (15j) define the UAV deployment range regarding spatial boundaries.

With coupled variables, mixed integers, and non-convex constraints, (P1) is a MINLP problem, making it complex and challenging to solve.

3. Proposed solution

In this section, we propose a low-complexity iterative optimization algorithm to solve (P1) by decomposing it into four manageable subproblems. This decomposition is grounded in the Block Coordinate Descent (BCD) method, which enables the iterative optimization of each subproblem while holding the others fixed. To ensure efficient optimization, each subproblem is carefully reformulated into a convex or tractable form, enabling faster convergence. The iterative process continues until the total latency converges below a predefined threshold. Although the decomposition may introduce minor discrepancies compared to the exact solution of (P1), the strong convergence of the BCD method ensures that these differences are negligible [19,35,36]. The overall framework of the proposed algorithm is illustrated in Fig. 3.

3.1. Task offloading decision optimization

Given **X**, **Q**, and **A**, the optimization problem with respect to task offloading decision **B** can be formulated as

$$(\text{P2}): \quad \min_{\{\mathbf{B},\varphi\}} \ \varphi \tag{16a}$$

$$\text{s.t.} \quad \sum_{k=1}^{K} \left(\beta_{k,0} T_k^{\text{loc}} + \sum_{m=1}^{M} \beta_{k,m} T_{k,m}^{\text{edge}} \right) \le \varphi, \tag{16b}$$

$$\sum_{k=1}^{K} \beta_{k,m} \chi_{k,m} \le 1, \forall m, \tag{16c}$$

$$\sum_{m=0}^{M} \beta_{k,m} = 1, \forall k, \tag{16d}$$

$$\sum_{k=1}^{K} \beta_{k,m} \le N_{\max}, \forall m, \tag{16e}$$

$$\beta_{k,m} \in \{0,1\}, \forall k, m. \tag{16f}$$

Since (P2) involves both continuous and binary variables, it represents a typical Mixed-Integer Linear Programming (MILP) problem. The solution space for the continuous variables forms a polyhedron, while the binary variables define a discrete set. The complexity of an MILP problem stems from the need to explore the polyhedral solution space of the continuous variables while simultaneously handling the combinatorial complexity of the binary variables, rendering it an NP-hard problem.

To address this complexity, we apply the linear relaxation method, relaxing the binary variables $\beta_{k,m}$ to continuous variables that can take any real value between 0 and 1, i.e., $0 \le \beta_{k,m} \le 1$. As a result, (P2) is reformulated as

$$(\text{P2.1}): \quad \min_{\{\mathbf{B},\varphi\}} \ \varphi \tag{17a}$$

$$\text{s.t.} \quad \text{(16b)–(16e)}, \tag{17b}$$

$$0 \le \beta_{k,m} \le 1, \forall k, m. \tag{17c}$$

By doing so, the original MILP problem is transformed into a continuous Linear Programming (LP) problem, which can be efficiently solved using standard LP solvers such as the Simplex or interior-point method [37]. Once the relaxed continuous solution is obtained, we verify whether it satisfies the binary constraints. If not, heuristic methods are employed to convert the solution back to binary form.

3.2. Bandwidth allocation optimization

Given **B**, **Q**, and **A**, the optimization problem with respect to bandwidth allocation **X** can be formulated as

$$(\text{P3}): \quad \min_{\{\mathbf{X},\varphi\}} \ \varphi \tag{18a}$$

$$\text{s.t.} \quad \sum_{k=1}^{K} \left(\beta_{k,0} T_k^{\text{loc}} + \sum_{m=1}^{M} \beta_{k,m} T_{k,m}^{\text{edge}} \right) \le \varphi, \tag{18b}$$

$$\sum_{k=1}^{K} \beta_{k,m} \chi_{k,m} \le 1, \forall m, \tag{18c}$$

$$\chi_{k,m} \in [0,1], \forall k, m. \tag{18d}$$

It can be observed that the objective function φ in (P3) is linear with respect to $\chi_{k,m}$, where T_k^{loc}, $T_{k,m}^{\text{edge}}$, and $\beta_{k,m}$ are constants. Moreover, all the constraints are linear inequalities with respect to $\chi_{k,m}$. Therefore, (P3) is a typical LP problem, ensuring the existence and uniqueness of the global optimal solution. Given this property, we solve (P3) using the interior-point method, which performs particularly well in handling large-scale LP problems, offering excellent global convergence properties and reliably converging to the optimal solution [38]. Furthermore, the theoretical time complexity of the interior-point method is polynomial, providing better performance guarantees compared to the worst-case exponential complexity of the Simplex method. Specifically, we begin by selecting an initial feasible point $\mathbf{X}^{(0)}$ that satisfies all constraints, and the solution is then updated iteratively using the following formula

$$\mathbf{X}^{(l+1)} = \mathbf{X}^{(l)} + \mu \Delta \mathbf{X}^{(l)}, \tag{19a}$$

$$\varphi^{(l+1)} = \varphi^{(l)} + \mu \Delta \varphi^{(l)}, \tag{19b}$$

where μ is the step size, and $\Delta X^{(l)}$ and $\Delta \varphi^{(l)}$ are the search directions in the current iteration. By satisfying appropriate convergence criteria, the interior-point method guarantees the accuracy of the solution, ultimately leading to the optimal solution for (P3).

3.3. 3D multi-UAV deployment optimization

Given **B**, **X**, and **A**, the optimization problem with respect to 3D multi-UAV deployment **Q** can be formulated as

$$(\text{P4}): \quad \min_{\{\mathbf{Q},\varphi\}} \ \varphi \tag{20a}$$

$$\text{s.t.} \quad \sum_{k=1}^{K} \left(\beta_{k,0} T_k^{\text{loc}} + \sum_{m=1}^{M} \beta_{k,m} T_{k,m}^{\text{edge}} \right) \le \varphi, \tag{20b}$$

$$X_{\min} \le x_m \le X_{\max}, \forall m, \tag{20c}$$

$$Y_{\min} \le y_m \le Y_{\max}, \forall m, \tag{20d}$$

$$H_{\min} \le h_m \le H_{\max}, \forall m. \tag{20e}$$

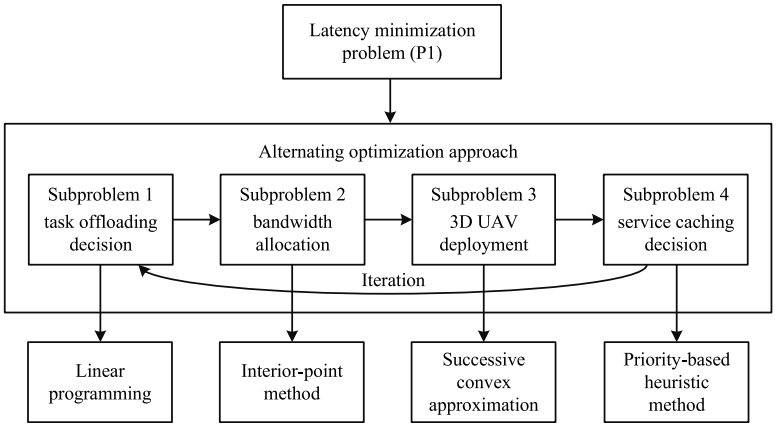


Fig. 3. Overall framework of the proposed algorithm.

In (P4), although constraints (20c)–(20e) are linear inequalities, the presence of non-convex terms such as $R_{k,m}$ and $R_{\mathrm{H},m}$ in the objective function (20a) and constraint (20b) makes (P4) a non-convex optimization problem. To address this, we apply the SCA technique, which iteratively approximates the non-convex components with convex functions, allowing the solution to progressively converge towards the optimal [39]. For clarity, let us define

$$x = \frac{p_k h_{k,m}}{\sigma^2}, \tag{21}$$

$$x^{(r)} = \frac{p_k h_{k,m}^{(r)}}{\sigma^2}. \tag{22}$$

In the rth iteration, the upper bound of $R_{k,m}$ is derived using the first-order Taylor expansion at the current point $h_{k,m}^{(r)}$, then we have

$$R_{k,m} \le R_{k,m}^{\mathrm{up}} = \chi_{k,m} B_{\mathrm{U}} \left(R_{k,m}^{(r)} + \nabla R_{k,m}^{(r)} \left(h_{k,m} - h_{k,m}^{(r)} \right) \right), \tag{23}$$

where

$$\begin{aligned} R_{k,m}^{(r)} &= \log_2 \left(1 + x^{(r)}\right) \\ &= \log_2 \left(1 + \frac{p_k h_{k,m}^{(r)}}{\sigma^2} \right), \end{aligned} \tag{24}$$

and

$$\begin{aligned} \nabla R_{k,m}^{(r)} &= \frac{p_k/\sigma^2}{\left(1 + x^{(r)}\right) \ln 2} \\ &= \frac{p_k/\sigma^2}{\left(1 + \frac{p_k h_{k,m}^{(r)}}{\sigma^2} \right) \ln 2}, \end{aligned} \tag{25}$$

where $\nabla R_{k,m}^{(r)}$ is the first-order derivative of $R_{k,m}$ with respect to $h_{k,m}^{(r)}$.

Similarly, for $R_{\mathrm{H},m}$, let

$$y = \frac{p_{\mathrm{H}} h_{\mathrm{H},m}}{\sigma^2}, \tag{26}$$

$$y^{(r)} = \frac{p_{\mathrm{H}} h_{\mathrm{H},m}^{(r)}}{\sigma^2}. \tag{27}$$

In the rth iteration, the upper bound of $R_{\mathrm{H},m}$ is derived using the first-order Taylor expansion at the current point $h_{\mathrm{H},m}^{(r)}$, given by

$$R_{\mathrm{H},m} \le R_{\mathrm{H},m}^{\mathrm{up}} = B_{\mathrm{H}} \left(R_{\mathrm{H},m}^{(r)} + \nabla R_{\mathrm{H},m}^{(r)} \left(h_{\mathrm{H},m} - h_{\mathrm{H},m}^{(r)} \right) \right), \tag{28}$$

where

$$\begin{aligned} R_{\mathrm{H},m}^{(r)} &= \log_2 \left(1 + y^{(r)}\right) \\ &= \log_2 \left(1 + \frac{p_{\mathrm{H}} h_{\mathrm{H},m}^{(r)}}{\sigma^2} \right), \end{aligned} \tag{29}$$

and

$$\begin{aligned} \nabla R_{\mathrm{H},m}^{(r)} &= \frac{p_{\mathrm{H}}/\sigma^2}{\left(1 + y^{(r)}\right) \ln 2} \\ &= \frac{p_{\mathrm{H}}/\sigma^2}{\left(1 + \frac{p_{\mathrm{H}} h_{\mathrm{H},m}^{(r)}}{\sigma^2} \right) \ln 2}, \end{aligned} \tag{30}$$

where $\nabla R_{\mathrm{H},m}^{(r)}$ is the first-order derivative of $R_{\mathrm{H},m}$ with respect to $h_{\mathrm{H},m}^{(r)}$.

By replacing $R_{k,m}$ and $R_{\mathrm{H},m}$ with their respective upper bounds $R_{k,m}^{\mathrm{up}}$ and $R_{\mathrm{H},m}^{\mathrm{up}}$, (P4) is reformulated as

$$(\mathrm{P4.1}): \quad \min_{\{\mathbf{Q},\varphi\}} \ \varphi \tag{31a}$$

$$\text{s.t.} \quad \sum_{k=1}^{K} \left(\begin{array}{l} \beta_{k,0} T_k^{\mathrm{loc}} + \\ \sum_{m=1}^{M} \beta_{k,m} \left(\frac{l_{k,i}}{R_{k,m}^{\mathrm{up}}} + t_{k,m}^{\mathrm{comp}} + \left(1 - \alpha_{m,i}\right) \frac{d_i}{R_{\mathrm{H},m}^{\mathrm{up}}} \right) \end{array} \right) \le \varphi, \tag{31b}$$

$$\text{(20c)–(20e)}. \tag{31c}$$

Following the above transformations, (P4.1) is converted into a standard convex optimization problem, which can be efficiently solved using optimization tools such as CVX.

3.4. Service caching decision optimization

Given **B**, **X**, and **Q**, the optimization problem with respect to service caching decision **A** can be formulated as

$$(\mathrm{P5}): \quad \min_{\{\mathbf{A},\varphi\}} \ \varphi \tag{32a}$$

$$\text{s.t.} \quad \sum_{m=1}^{M} \beta_{k,m} \left(1 - \alpha_{m,i}\right) t_{\mathrm{H},m}^{\mathrm{down}} + \varpi_k \le \varphi, \tag{32b}$$

$$\sum_{i=1}^{I} \alpha_{m,i} d_i \le D_m, \forall m, \tag{32c}$$

$$\alpha_{m,i} \in \{0,1\}, \forall m, i, \tag{32d}$$

where $\varpi_k = \beta_{k,0} T_k^{\mathrm{loc}} + \sum_{m=1}^{M} \beta_{k,m} \left(t_{k,m}^{\mathrm{off}} + t_{k,m}^{\mathrm{comp}} \right)$. Since both the objective function (32a) and constraint (32b) contain nonlinear terms, (P5) remains a Mixed-Integer Nonlinear Programming (MINLP) problem.

The solution is typically obtained through the following steps: (1) Relax the binary variables to continuous values, i.e., replace constraint (32d) with $0 \leq \alpha_{m,i} \leq 1$; (2) Solve the reformulated problem using convex optimization methods or the concave-convex procedure; and (3) Convert the continuous variables back to binary integers. However, this approach is inefficient for large-scale problems [15]. In this subsection, we propose an efficient priority-based method to determine the service caching decisions, with a polynomial computational complexity of $\mathcal{O}(KM + K\log K + MI)$.

Ideally, assuming UAVs have sufficient cache capacity to store all services, i.e., $\sum_{i=1}^{I} \alpha_{m,i} d_i \leq D_m$, the optimal caching strategy would be to cache all services, i.e., $\alpha_{m,i} = 1, \forall m, i$. This ensures that all services are cached on the UAVs, eliminating the need to download from the HAP, thereby achieving lower task latency. However, in most cases, UAVs have limited cache capacity and cannot store all services, i.e., $\sum_{i=1}^{I} \alpha_{m,i} d_i > D_m$. It is straightforward to conclude that priority should be given to caching the services required by tasks that contribute the most to minimizing φ. In other words, to better utilize the cache space, services for tasks with the highest $\varpi_k + t_{\mathrm{H},m}^{\mathrm{down}}$ values should be cached on the UAVs. Let $\mathcal{K}_m = \{k \in \mathcal{K} | \beta_{k,m}\}$ and $|\mathcal{K}_m|$ denote the set of TDs associated with UAV m and their number, respectively. Similarly, let $\mathcal{I}_k \in \mathcal{I}$ and $|\mathcal{I}_k|$ represent the set and number of services required by the tasks of these TDs in $\mathcal{K}_m$. We then sort the elements in $\mathcal{I}_k$ in descending order based on their $\varpi_k + t_{\mathrm{H},m}^{\mathrm{down}}$ values.

For UAV m, let the remaining cache capacity be denoted as $R_m = D_m$. Based on the sorting results, high-priority services are cached on UAV m until its capacity is fully utilized. The update formula for $\alpha_{m,i}$ is given by

$$\alpha_{m,i}^{*} = \begin{cases} 1, & \text{if} \quad \sum_{i=1}^{I} \alpha_{m,i} d_i \leq R_m \\ 0, & \text{otherwise} \end{cases}, \tag{33}$$

where

$$R_m = R_m - \sum_{i=1}^{I} \alpha_{m,i} d_i. \tag{34}$$

4. Complexity and convergence analysis

In this section, we analyze the complexity and convergence of the proposed algorithm, which iteratively solves four subproblems, refining the solution with each step. Although some approximations are introduced, the iterative updates ensure that the solution converges to a near-optimal result, with only a minimal deviation from the global optimum. Algorithm 1 outlines the iterative process used to solve (P1).

4.1. Complexity analysis

In (P2), we solve the linear relaxation using an LP solver, resulting in a computational complexity of $\mathcal{O}(KM)$. For (P3), the interior-point method is employed, which has a computational complexity of $\mathcal{O}\left((KM)^3\right)$. In (P4), the SCA technique transforms the non-convex problem into a convex optimization problem, which is solved using convex optimization tools with a complexity of $\mathcal{O}\left(M^{3.5}\right)$. The computational complexity of (P5) is $\mathcal{O}(KM + K\log K + MI)$.

Although some subproblems, particularly (P3) and (P4), exhibit higher computational complexity due to their cubic and super-cubic terms, it is important to note that the overall complexity of the proposed algorithm remains polynomial. The polynomial nature of the complexity ensures that, even for large-scale systems, the algorithm is computationally feasible. Thus, the total complexity of the iterative algorithm is $\mathcal{O}(R(2KM + (KM)^3 + M^{3.5} + K\log K + MI))$, where R represents the total number of iterations.

Algorithm 1: Iterative Optimization Algorithm for Solving (P1)

Input: $K, M, I, D_m, L_{k,i}, \mathbf{w}_k, \mathbf{w}_{\mathrm{H}}, \beta_0, K_{k,m}, f_c, B_{\mathrm{U}}, B_{\mathrm{H}}, p_k, p_{\mathrm{H}}, \sigma^2, N_{\max}$, and the convergence threshold $\varepsilon > 0$.

Initialize: Slack variables, iteration index $r = 0$, and initial feasible points $\mathbf{B}^{(0)}$, $\mathbf{X}^{(0)}$, $\mathbf{Q}^{(0)}$, and $\mathbf{A}^{(0)}$.

repeat
- calculate $\mathbf{B}^{(r+1)}$ for (P2.1) with given $\mathbf{X}^{(r)}$, $\mathbf{Q}^{(r)}$, and $\mathbf{A}^{(r)}$;
- calculate $\mathbf{X}^{(r+1)}$ for (P3) with given $\mathbf{B}^{(r+1)}$, $\mathbf{Q}^{(r)}$, and $\mathbf{A}^{(r)}$;
- calculate $\mathbf{Q}^{(r+1)}$ for (P4.1) with given $\mathbf{B}^{(r+1)}$, $\mathbf{X}^{(r+1)}$, and $\mathbf{A}^{(r)}$;
- calculate $\mathbf{A}^{(r+1)}$ for (P5) with given $\mathbf{B}^{(r+1)}$, $\mathbf{X}^{(r+1)}$, and $\mathbf{Q}^{(r+1)}$;
- Update $r \leftarrow r + 1$;

until $T_{\mathrm{total}}^{(r+1)} - T_{\mathrm{total}}^{(r)} \leq \varepsilon$;

Output: The optimized solutions $\mathbf{B}^{*}$, $\mathbf{X}^{*}$, $\mathbf{Q}^{*}$, and $\mathbf{A}^{*}$.

4.2. Convergence analysis

Let $T_{\mathrm{total}}(\mathbf{B}, \mathbf{X}, \mathbf{Q}, \mathbf{A})$ denote the objective function value of the formulated joint optimization problem. After r iterations of the algorithm, given $\mathbf{X}^{(r)}$, $\mathbf{Q}^{(r)}$, and $\mathbf{A}^{(r)}$, solving (P2.1), (P3), (P4.1), and (P5) results in

$$T_{\mathrm{total}}\left(\mathbf{B}^{(r)}, \mathbf{X}^{(r)}, \mathbf{Q}^{(r)}, \mathbf{A}^{(r)}\right) \leq T_{\mathrm{total}}\left(\mathbf{B}^{(r+1)}, \mathbf{X}^{(r)}, \mathbf{Q}^{(r)}, \mathbf{A}^{(r)}\right) \tag{35a}$$

$$T_{\mathrm{total}}\left(\mathbf{B}^{(r+1)}, \mathbf{X}^{(r)}, \mathbf{Q}^{(r)}, \mathbf{A}^{(r)}\right) \leq T_{\mathrm{total}}\left(\mathbf{B}^{(r+1)}, \mathbf{X}^{(r+1)}, \mathbf{Q}^{(r)}, \mathbf{A}^{(r)}\right) \tag{35b}$$

$$T_{\mathrm{total}}\left(\mathbf{B}^{(r+1)}, \mathbf{X}^{(r+1)}, \mathbf{Q}^{(r)}, \mathbf{A}^{(r)}\right) \leq T_{\mathrm{total}}\left(\mathbf{B}^{(r+1)}, \mathbf{X}^{(r+1)}, \mathbf{Q}^{(r+1)}, \mathbf{A}^{(r)}\right) \tag{35c}$$

$$T_{\mathrm{total}}\left(\mathbf{B}^{(r+1)}, \mathbf{X}^{(r+1)}, \mathbf{Q}^{(r+1)}, \mathbf{A}^{(r)}\right) \leq T_{\mathrm{total}}\left(\mathbf{B}^{(r+1)}, \mathbf{X}^{(r+1)}, \mathbf{Q}^{(r+1)}, \mathbf{A}^{(r+1)}\right) \tag{35d}$$

This analysis shows that the objective function value is non-increasing after each iteration. Given that the objective function is bounded below, the monotonic convergence ensures that the objective function value converges to a limit [6]. Therefore, the proposed iterative optimization algorithm is guaranteed to converge.

5. Simulation results and discussions

5.1. Simulation setup

The simulation models a dual-layer aerial MEC system consisting of one HAP and multiple UAVs, which provide computation and caching services to terrestrial TDs. The HAP is positioned at a fixed altitude of 20 km and is the central data hub for storing services [10]. Four UAVs are strategically deployed at altitudes between 200 and 400 m [11]. Each UAV has a cache capacity of 200 MB, allowing it to store a subset of services, with service data sizes ranging from 10 MB to 50 MB [15,16]. The simulation spans a 400×400 m^2 area, with 40 TDs randomly distributed. Each TD generates computation-intensive tasks with data sizes ranging from 1 MB to 5 MB, and the total CPU cycles required to complete each task range from 1000 to 3000, which is typical for IoT applications [18,20]. The local computation capacity of each TD is 500 MHz, while UAVs allocate 2 GHz of their computation resources to process offloaded tasks from the TDs [17]. The network operates with a 1 to 3 MHz bandwidth for T2U links and 5 to 10 MHz for H2U links. TDs transmit data to the UAVs using a power of 0.1 W, while UAVs communicate with the HAP at a higher power of 3 W to ensure stable transmission over long distances [23,26]. The path loss model assumes a channel power gain of −50 dB at a reference distance of 1 m, and the total directional antenna gain is set to 15 dBi. The noise power at the receivers is set to −168 dBm/Hz. Each UAV can handle up to 12 concurrent tasks from the TDs. The simulation parameters used in this paper are based on industry standards and previous research [39,40], as listed in Table 2 unless otherwise specified. The simulations were performed using MATLAB R2021b on a workstation with an Intel Core i7-10700 CPU and 16 GB of RAM.

Table 2
Simulation parameters.

Notations	Values	Notations	Values
H_{H}	20 km	M	4
$H_{\min}$	200 m	K	40
$H_{\max}$	400 m	I	50
$l_{k,i}$	[1, 5] MB	p_k	0.1 W
$c_{k,i}$	[1000, 3000] cycles	p_{H}	3 W
D_m	200 MB	f_k^{loc}	500 MHz
d_i	[10, 50] MB	$f_{k,m}$	2 GHz
B_{U}	[1, 3] MHz	σ^2	−168 dBm/Hz
B_{H}	[5, 10] MHz	G_0	15 dBi
$N_{\max}$	12	β_0	−50 dB

To comprehensively evaluate the performance of our proposed scheme, we compare it against four baseline schemes as follows.

- **Scheme 1: Nearest offloading.** Each TD offloads its task to the nearest UAV, regardless of whether it has cached the required service. If the service is not cached, the UAV retrieves it from the HAP before executing.
- **Scheme 2: UAV without caching.** UAVs do not have caching capability. Thus, all tasks, regardless of size, must be processed by retrieving the required service from the HAP before computation can begin.
- **Scheme 3: K-Means UAV deployment.** UAVs are deployed based on K-Means clustering results. Service caching, task offloading decisions, and bandwidth allocation are still optimized within this deployment strategy.
- **Scheme 4: Equal bandwidth allocation.** The bandwidth allocated by each UAV to the TDs it serves is equally and statically divided, without accounting for the dynamic variation in bandwidth demand.
- **Proposed joint optimization.** Service caching decisions, task offloading decisions, 3D UAV deployment, and bandwidth allocation are jointly optimized in this scheme.

5.2. Simulation initialization

Initially, no services are cached on any UAV, with all caching decision variables set to zero, i.e., $\alpha_{m,i} = 0$. The task offloading decisions are initialized by assigning each TD to offload its computation task to the nearest UAV, determined by the Euclidean distance between the UAVs and the TDs in their initial locations. Bandwidth is evenly distributed among the TDs, with each TD initially allocated an equal share of the available bandwidth. The horizontal coordinates of the UAVs are uniformly distributed across the area, and the initial altitude of each UAV is set to the maximum permissible height. The convergence threshold is set to $\epsilon = 10^{-3}$, meaning the iteration process terminates when the reduction in total system latency is smaller than this value.

5.3. Simulation results and performance analysis

Table 3 compares the run times of different algorithms under small and large system scales. The nearest offloading algorithm, which offloads tasks to the closest UAV without complex decisions, has relatively low computational complexity, with run times of 50 s for small scale and 68 s for large scale due to increased task volume. The UAV without caching algorithm retrieves data from the HAP but avoids caching decisions, keeping complexity low, with run times of 57 s and 70 s, respectively. K-means deployment adds complexity by requiring clustering before resource allocation, resulting in longer run times of 84 s for small scale and 102 s for large scale. Though static in bandwidth division, the equal bandwidth allocation algorithm still involves task offloading and caching decisions, leading to 60 s and 73 s run times. With its multi-variable joint optimization, the proposed algorithm requires the most computation time, running at 112 s for small scale and 136 s for large scale, but this is acceptable given its significant performance improvements.

Table 3
Comparison of runtime for different algorithms under various system scales (small scale: 40 TDs, 4 UAVs; large scale: 50 TDs, 5 UAVs).

Algorithm	Runtime (s) for small scale	Runtime (s) for large scale
Nearest offloading	50	68
UAV without caching	57	70
K-means UAV deployment	84	102
Equal bandwidth allocation	60	73
Proposed algorithm	112	136

Fig. 4 illustrates the optimized multi-UAV deployment. As shown in Fig. 4(a), UAVs are distributed widely in the horizontal plane, effectively minimizing the transmission distance between TDs and UAVs, thereby reducing communication latency. The UAVs are concentrated in high-density user areas, highlighting the algorithm's ability to adapt to hotspots. Fig. 4(b) depicts the UAVs' distribution at varying altitudes, fully utilizing the 3D space and balancing the trade-off between task offloading and service downloading. Lower altitude UAVs are positioned closer to the TDs, reducing uplink transmission latency, while higher altitude UAVs are positioned nearer the HAP, minimizing downlink latency. This 3D deployment strategy not only mitigates signal blockages but also improves transmission efficiency.

Fig. 5 verifies the convergence of the proposed scheme. As the number of iterations increases, the total latency of both schemes gradually decreases and converges after a finite number of iterations. Specifically, the total latency of the random UAV deployment scheme decreases from 248 s to 150 s, while the total latency of the proposed scheme decreases from 210 s in the initial iteration to a stable 120 s. Compared to the random UAV deployment scheme, the proposed scheme achieves a 42.86% reduction in total latency. This improvement is attributed to the proposed scheme optimizing 3D multi-UAV deployment based on user demands and network conditions, allowing UAVs to be precisely positioned at optimal service locations, ensuring high efficiency and stability of communication links.

Fig. 6 illustrates the total latency versus iterations for different schemes. Initially, the total latency for all schemes is around 210 s. As the number of iterations increases, the total latency gradually decreases. The proposed scheme converges the fastest, reaching stability by the third iteration. Additionally, the proposed scheme achieves the lowest total latency, indicating that the joint optimization of all variables is the most effective approach. Moreover, optimizing task offloading and bandwidth allocation proves to be the most effective single-variable optimization strategy. This is because task offloading decisions directly impact both computation and transmission latency, while bandwidth allocation determines the efficiency of communication links. Although UAV deployment and service caching also contribute to latency optimization, their relative impact is smaller.

Fig. 7 shows the total latency versus iterations for different numbers of UAVs. As the iterations increase, the total latency decreases significantly for both configurations. However, the reduction is faster when $M = 5$. By the second iteration, the total latency for $M = 5$ drops to around 128 s, whereas it remains close to 150 s for $M = 4$. This indicates that increasing the number of UAVs accelerates the convergence process. Eventually, the total latency stabilizes at approximately 103 s after the fourth iteration for $M = 5$, compared to around 120 s for $M = 4$. This demonstrates the advantage of having more UAVs, as additional UAVs provide increased communication, computation, and caching resources, thereby effectively distributing the computation tasks and reducing total latency.

Fig. 8 shows the total latency versus iterations with $M = 5$, while Fig. 6 shows the scenario when $M = 4$, both with $K = 40$. By comparing their results, several conclusions can be drawn. Firstly,

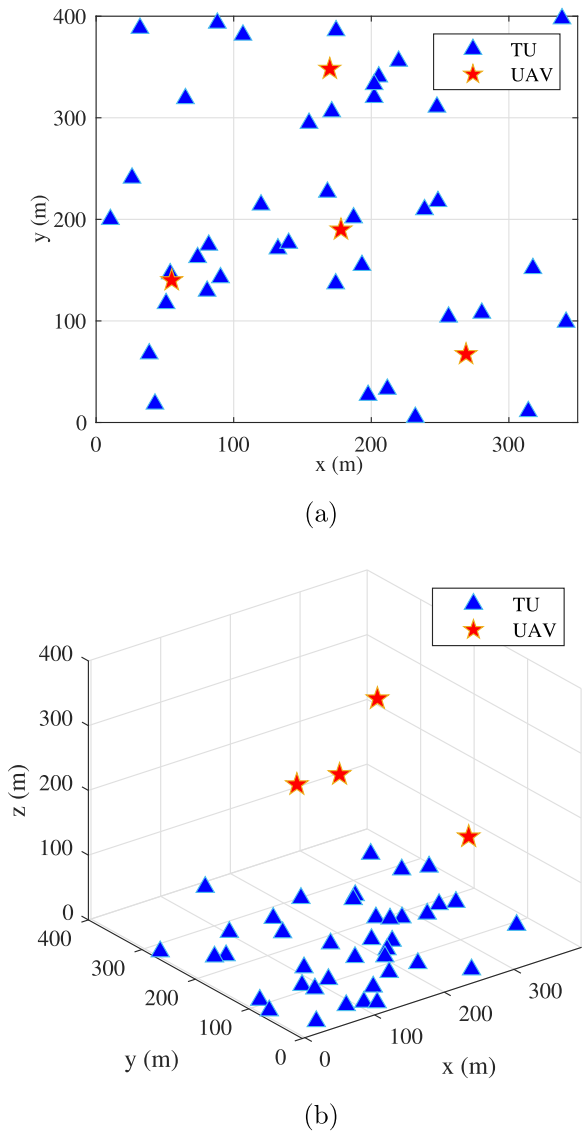


Fig. 4. Optimized multi-UAV deployment. (a) 2D multi-UAV deployment. (b) 3D multi-UAV deployment.

regardless of whether $M = 5$ or $M = 4$, the initial total latency for all schemes is relatively high. As the number of iterations increases, the total latency decreases significantly, with the proposed scheme demonstrating a faster convergence speed. Secondly, the final total latency of the proposed scheme with $M = 5$ is 103 s, compared to 120 s with $M = 4$, representing a 14.17% reduction in total latency. This indicates that increasing the number of UAVs leverages the advantages of multi-UAV collaboration in distributing computation and caching tasks, significantly improving the optimization results.

Fig. 9 shows the total latency versus iterations for different numbers of TDs. The results indicate that as the number of iterations increases, the total latency significantly decreases in all scenarios. Initially, when $K = 50$, the total latency is higher, approximately 293 s, whereas with $K = 40$, the total latency is 210 s. By the fifth iteration, the total latency stabilizes at around 206 s for $K = 50$ and 120 s for $K = 40$. The results clearly show that a higher number of TDs results in greater total latency. This is because an increased number of TDs amplifies the computation and communication load, leading to intensified resource

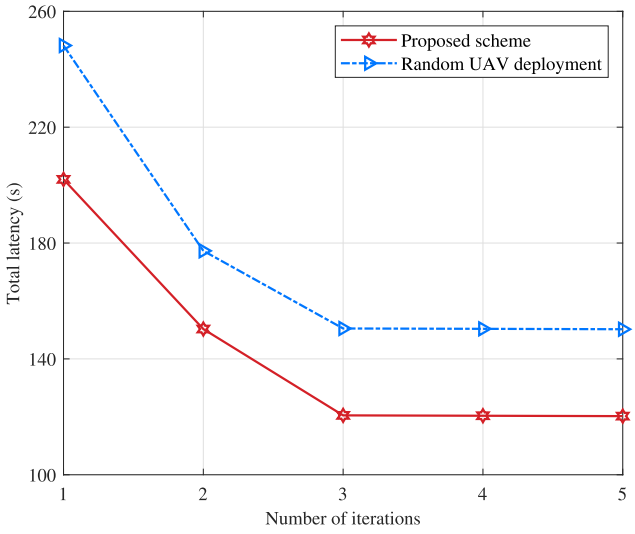


Fig. 5. Total latency vs. iterations: Proposed scheme vs. Random UAV deployment.

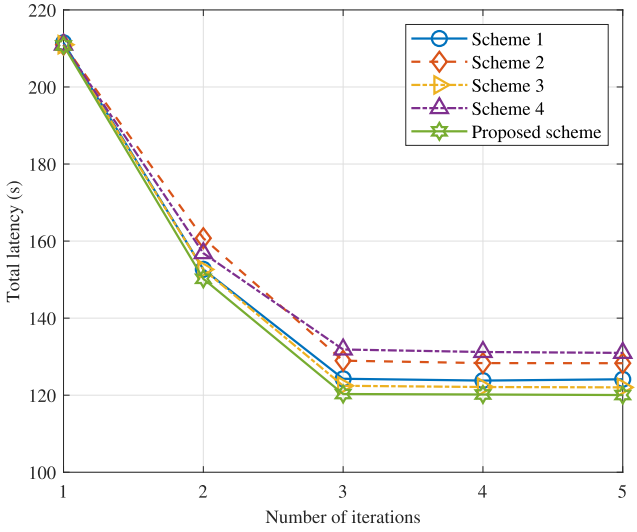


Fig. 6. Total latency vs. iterations for different schemes.

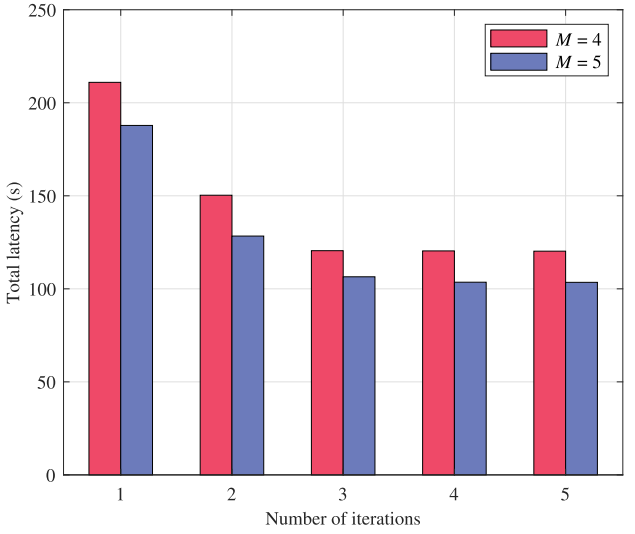


Fig. 7. Total latency vs. iterations for different numbers of UAVs.

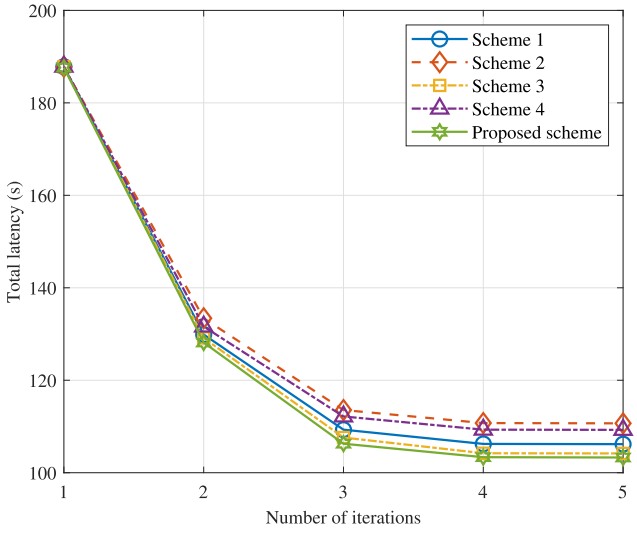


Fig. 8. Total latency vs. iterations with $M = 5$.

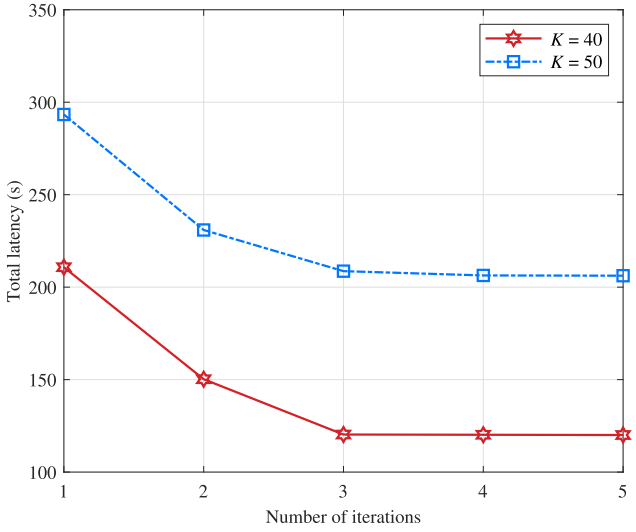


Fig. 9. Total latency vs. iterations for different numbers of TDs.

competition and increased scheduling complexity, thereby extending the task processing time. These findings highlight the importance of efficient resource management and scheduling to mitigate the impact of high TD density on system performance.

Fig. 10 shows the total latency versus iterations with $K = 50$, while Fig. 6 shows the scenario when $K = 40$, both with $M = 4$. Firstly, in the initial iteration, the total latency with $K = 50$ is significantly higher than with $K = 40$, indicating that as the number of TDs increases, the initial load and latency increase significantly. In the second iteration, the total latency for the proposed scheme decreases rapidly in both scenarios, but the decrease is more pronounced with $K = 50$, demonstrating the effectiveness of the algorithm in optimizing system performance under high load conditions. After the third iteration, the total latency stabilizes. For the $K = 50$ scenario, the total latency of the proposed scheme decreases from 193 s to 209 s, a reduction of 29.69%; for the $K = 40$ scenario, the total latency decreases from 211 s to 120 s, a reduction of 42.86%. This indicates that while the proposed

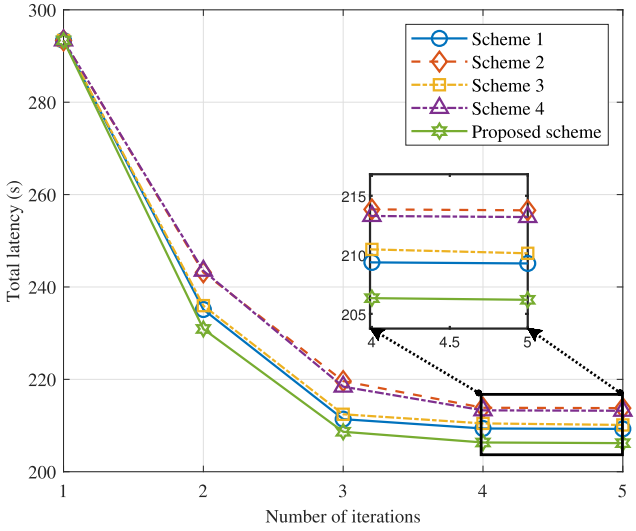


Fig. 10. Total latency vs. iterations with $K = 50$.

scheme has a more significant optimization effect with fewer TDs, it still maintains good performance with a larger number of TDs.

6. Conclusion

In this paper, we propose a novel dual-layer aerial MEC architecture that integrates multiple UAVs and one HAP to provide computation and caching services for resource-constrained TDs. The joint optimization of service caching, task offloading, 3D UAV deployment, and bandwidth allocation is formulated as a complex MINLP problem. To solve this, we develop an efficient iterative optimization algorithm based on linear relaxation and SCA techniques. The system scalability was validated under various configurations, with the proposed scheme consistently outperforming other baselines in both small-scale and large-scale scenarios. Although the proposed algorithm incurred slightly longer runtimes than other baselines, the substantial performance improvements make this trade-off acceptable. Among the variables, task offloading and bandwidth allocation emerge as the most effective for single-variable optimization, while increasing the number of UAVs to a certain extent further enhances system performance. Despite the promising results, this work assumes static UAV deployment, which may limit its applicability in dynamic environments.

Future research should focus on optimizing UAV trajectory to accommodate mobile users and varying task demands. Integrating energy-efficient strategies, such as wireless energy harvesting, will be key to extending the operational lifespan of UAVs and IoT devices. Additionally, task queuing and prioritization mechanisms should be developed to ensure timely processing of critical tasks. Incorporating machine learning techniques will enhance system efficiency in complex, large-scale environments.

CRediT authorship contribution statement

Yue Zhang: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation. **Zhenyu Na:** Supervision, Funding acquisition. **Zihao Wen:** Writing – review & editing. **Arumugam Nallanathan:** Supervision. **Weidang Lu:** Writing – review & editing, Supervision, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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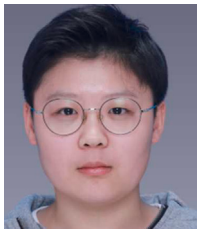
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